

2.5 Curvature Worksheet

1. Which of the following expressions is equal to the curvature of a curve given by the parameterization $\mathbf{r}(t)$ where $t \geq 0$, $\mathbf{T}(t)$ is the unit tangent vector, and $s(t) = \int_0^t \|\mathbf{r}'(u)\| du$? Simply circle your answer.

You do not need to provide justification.

A. $\left\| \frac{d\mathbf{T}}{dt} \right\|$ B. $\frac{d\mathbf{T}}{ds}$ C. $\frac{d\mathbf{T}}{dt}$ D. $\left\| \frac{d\mathbf{T}}{ds} \right\|$

The correct answer is **D**.

2. Consider the curve $\mathbf{r}(t) = \langle t^2, \ln(t), t \ln(t) \rangle$. Find the curvature of $\mathbf{r}(t)$ at the point $(1, 0, 0)$.

We know that $\kappa(t) = \frac{\|\mathbf{r}'(t) \times \mathbf{r}''(t)\|}{\|\mathbf{r}'(t)\|^3}$ and that $\mathbf{r}(t)$ reaches the point $(1, 0, 0)$ when $t = 1$.

Therefore, we want to find $\kappa(1) = \frac{\|\mathbf{r}'(1) \times \mathbf{r}''(1)\|}{\|\mathbf{r}'(1)\|^3}$

We calculate:

$$\mathbf{r}'(t) = \left\langle 2t, \frac{1}{t}, \ln(t) + 1 \right\rangle \rightarrow \mathbf{r}'(1) = \langle 2, 1, 1 \rangle$$

$$\mathbf{r}''(t) = \left\langle 2, -\frac{1}{t^2}, \frac{1}{t} \right\rangle \rightarrow \mathbf{r}''(1) = \langle 2, -1, 1 \rangle$$

$$\|\mathbf{r}'(1)\| = \sqrt{4 + 1 + 1} = \sqrt{6}$$

$$\mathbf{r}'(1) \times \mathbf{r}''(1) = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 2 & 1 & 1 \\ 2 & -1 & 1 \end{vmatrix} = \langle 2, 0, -4 \rangle$$

$$\|\mathbf{r}'(1) \times \mathbf{r}''(1)\| = \sqrt{4 + 0 + 16} = \sqrt{20}$$

$$\text{Therefore, } \kappa(1) = \frac{\|\mathbf{r}'(1) \times \mathbf{r}''(1)\|}{\|\mathbf{r}'(1)\|^3} = \frac{\sqrt{20}}{(\sqrt{6})^3} = \frac{2\sqrt{5}}{6\sqrt{6}} = \frac{1}{3} \sqrt{\frac{5}{6}}$$

3. Consider the curve $\mathbf{r}(t) = \langle \cos(t), \sin(t), t \rangle$.

a) Determine a formula to measure the curvature of $\mathbf{r}(t)$ at any value of t .

$$\text{We know that } \kappa(t) = \frac{\|\mathbf{r}'(t) \times \mathbf{r}''(t)\|}{\|\mathbf{r}'(t)\|^3}.$$

We calculate:

$$\mathbf{r}'(t) = \langle -\sin(t), \cos(t), 1 \rangle$$

$$\mathbf{r}''(t) = \langle -\cos(t), -\sin(t), 0 \rangle$$

$$\|\mathbf{r}'(t)\| = \sqrt{\sin^2(t) + \cos^2(t) + 1} = \sqrt{2}$$

$$\mathbf{r}'(t) \times \mathbf{r}''(t) = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ -\sin(t) & \cos(t) & 1 \\ -\cos(t) & -\sin(t) & 0 \end{vmatrix} = \langle \sin(t), \cos(t), \sin^2(t) + \cos^2(t) \rangle = \langle \sin(t), \cos(t), 1 \rangle$$

$$\|\mathbf{r}'(t) \times \mathbf{r}''(t)\| = \sqrt{\sin^2(t) + \cos^2(t) + 1} = \sqrt{2}$$

$$\text{Therefore, } \kappa(t) = \frac{\|\mathbf{r}'(t) \times \mathbf{r}''(t)\|}{\|\mathbf{r}'(t)\|^3} = \frac{\sqrt{2}}{(\sqrt{2})^3} = \frac{1}{2}$$

b) Determine the curvature of $\mathbf{r}(t)$ at the point $(1, 0, 2\pi)$, $(1, 0, 0)$, and $(-1, 0, \pi)$.

We see from part (a) that for any value of t the curvature is always $\kappa(t) = \frac{1}{2}$. Therefore the curvature at all of these points is $\frac{1}{2}$.

4. Consider the curve $\mathbf{r}(t) = \langle t^2, 0, t \rangle$.

a) Determine a formula to measure the curvature of $\mathbf{r}(t)$ at any value of t .

$$\text{We know that } \kappa(t) = \frac{\|\mathbf{r}'(t) \times \mathbf{r}''(t)\|}{\|\mathbf{r}'(t)\|^3}.$$

We calculate:

$$\mathbf{r}'(t) = \langle 2t, 0, 1 \rangle \quad \& \quad \mathbf{r}''(t) = \langle 2, 0, 0 \rangle$$

$$\|\mathbf{r}'(t)\| = \sqrt{4t^2 + 0 + 1} = \sqrt{4t^2 + 1}$$

$$\mathbf{r}'(t) \times \mathbf{r}''(t) = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 2t & 0 & 1 \\ 2 & 0 & 0 \end{vmatrix} = \langle 0, 2, 0 \rangle$$

$$\|\mathbf{r}'(t) \times \mathbf{r}''(t)\| = \sqrt{0 + 4 + 0} = 2$$

$$\text{Therefore, } \kappa(t) = \frac{\|\mathbf{r}'(t) \times \mathbf{r}''(t)\|}{\|\mathbf{r}'(t)\|^3} = \frac{2}{(\sqrt{4t^2 + 1})^3} = \frac{2}{(4t^2 + 1)^{3/2}}$$

b) Determine the curvature of $\mathbf{r}(t)$ at the points $(0, 0, 0)$, $(1, 0, 1)$, and $(4, 0, 2)$.

We know that $\mathbf{r}(t)$ arrives at:

$$(0, 0, 0) \text{ when } t = 0$$

$$(1, 0, 1) \text{ when } t = 1$$

$$(4, 0, 2) \text{ when } t = 2$$

Therefore, the curvature at each point is as follows:

$$\text{At } (0, 0, 0), \kappa(0) = \frac{2}{(4(0)^2 + 1)^{3/2}} = 2.$$

$$\text{At } (1, 0, 1), \kappa(1) = \frac{2}{(4(1)^2 + 1)^{3/2}} = \frac{2}{5^{3/2}}.$$

$$\text{At } (4, 0, 2), \kappa(2) = \frac{2}{(4(2)^2 + 1)^{3/2}} = \frac{2}{17^{3/2}}.$$